- *The queue acts as a buffer*. Producers and consumers don't have to run at the same speed. For example, you can add a batch of 1,000 messages in one minute while your consumers always process 10 messages per second. Sooner or later, the consumers will catch up, and the queue will be empty again.
- *The queue hides your backend*. Similar to the load balancer, message producers have no knowledge of the consumers. You can even stop all consumers and still produce messages. This is handy while doing maintenance on your consumers.

When decoupled, the producers and consumers don't know each other; they both only know about the message queue. Figure 14.4 illustrates this principle.

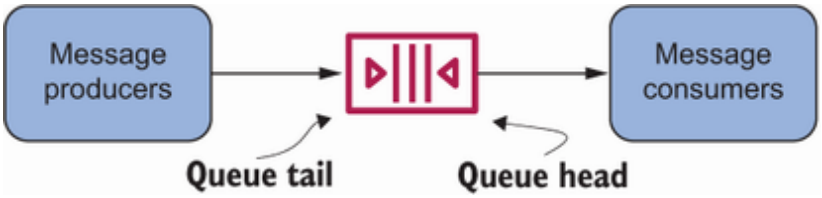


Figure 14.4 Producers send messages to a message queue while consumers read messages.

As you decoupled the sender from the receiver, the sender could even put new messages into the queue while no one is consuming messages, with the message queue acting as a buffer. To prevent message queues from growing infinitely large, messages are saved for only a certain amount of time. If you consume a message from a message queue, you must acknowledge the successful processing of the message to permanently delete it from the queue.

How do you implement asynchronous decoupling on AWS? That's where the Simple Queue Service (SQS) comes into play. SQS offers